Stolen Notes: Determinant and Trace

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(stolen from Gowtham Kurri)*

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This is an early release; may contain a few errors

Here, unless explicitly stated otherwise, assume the matrices are square.

1 In the syllabus

1.1 Determinant—Axiomatic Definition

- Usually determinant is thought of as a quantity
  - that can be computed for a matrix and
  - is usually presented as a formula.
- Here, we take a different approach.
  - We start by listing three desirable properties that we want,
    - a function $\det : \mathbb{F}^{n\times n} \to \mathbb{F}$ to satisfy,
    - where $\mathbb{F}$ is the field of real numbers or that of complex numbers.
  - We postulate these as axioms and say that
    - any function satisfying these axioms, will be called a "determinant".
  - Later, we argue that such a function is *unique*
    - leading us to the well-defined formula for *the* determinant.

Definition 1 (Axioms for a Determinant)**.** Consider any function $\det : \mathbb{F}^{n\times n} \to \mathbb{F}$ (that we sometimes also denote by $|\cdot|$ where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ (real or complex).

We say that det is a *determinant* if it satisfies the following properties:

*Almost entirely based on Gowtham Kurri's handwritten note on the same topic (who collected these results from a variety of different sources, including Gilbert Strang's lectures/text and the textbook 'Linear Algebra Done Wrong'). I re-arranged a few things and (hopefully) improved the presentation/arguments slightly for my own understanding. If spot any errors, please drop me an email. Thanks.

1. *Normalisation.*

   We require that $\det I = 1$.

2. *Anti-symmetry.*

   Let $A \in \mathbb{F}^{(n\times n)}$ and A' be the same as A except two of its rows have been exchanged.

   We require that $\det(A) = \det(A')$.

3. *Linearity in each row.*

   Determinant is linear in each row separately.

   Let $A \in \mathbb{F}^{n\times n}$ and denote by v_j the jth row of A, i.e.

$$A = \begin{bmatrix} — & v_1 & — \\ & \vdots & \\ — & v_n & — \end{bmatrix}.$$

   Define the notation $\det(v_1, v_2, \dots v_n) := \det(A)$.

   We require that

$$\det(v_1, \dots, v_{k-1}, \quad \underbrace{c_1 v_k + c_2 u_k}_{k\text{th row}}, \quad v_{k+1}, \dots v_n)$$
$$=$$
$$c_1 \det(v_1, \dots v_{k-1}, \quad \underbrace{v_k}_{k\text{th row}}, \quad v_{k+1}, \dots v_n) \quad +$$
$$c_2 \det(v_1, \dots v_{k-1}, \quad \underbrace{u_k}_{k\text{th row}}, \quad v_{k+1}, \dots v_n).$$

- Let us quickly recall the formulae for the determinant
  - that we learnt in high-school, for 2×2 and 3×3 determinants:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc,$$

    and

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei - afh + bfg - bdi + cdh - ceg.$$

- To ensure we have understood Definition 1,
  - let us check that the three properties are satisfied
    - by our formula for the 2×2 and 3×3 case.

- Normalisation:
  - Clearly, $\begin{vmatrix} 1 & \\ & 1 \end{vmatrix} = 1$ (set $b = c = 0$, ad survives).
  - Similarly, $\begin{vmatrix} 1 & & \\ & 1 & \\ & & 1 \end{vmatrix} = 1$ (set $b = c = d = f = g = h = 0$; only the aei term survives).
- Antisymmetry:
  - Clearly, $\begin{vmatrix} 1 & \\ & 1 \end{vmatrix} = 1$ but $\begin{vmatrix} & 1 \\ 1 & \end{vmatrix} = -1$ (only $-bc$ survives).
  - Again, $\begin{vmatrix} 1 & & \\ & 1 & \\ & & 1 \end{vmatrix} = 1$ but $\begin{vmatrix} 1 & & \\ & & 1 \\ & 1 & \end{vmatrix} = 1 \cdot (-1) = -1$.
- Linearity in each row:
  - We leave this as a small exercise to verify that

$$\begin{vmatrix} a + a' & b + b' \\ c & d \end{vmatrix} \overset{\text{exercise}}{=} \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}.$$

  - This is immediate

$$\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix}.$$

- It is worth pointing out that, in general,

$$\det(A + B) \neq \det(A) + \det(B)$$

  and similarly, in general,

$$\det(tA) \neq t \det(A).$$

- In fact, it turns out that if A is an $n \times n$ matrix, then

$$\det(tA) = t^n \det(A).$$

1.2 Properties of the determinant

Before we show that these three axioms uniquely specify the determinant function, we derive some interesting properties, assuming them (the three axioms).

Lemma 2. *Let* $\det : \mathbb{F}^{n \times n} \to \mathbb{F}$ *be as in Definition 1, satisfying the three axioms listed there. Let* $A \in \mathbb{F}^{n \times n}$ *be any . Then,* $\det$ *also satisfies the following properties.*

1. *If any two rows of A are equal, then* $\det A = 0$.

2. *If A has a zero row, then* $\det A = 0$.

3. *If the rows of A are linearly dependent, then* $\det A = 0$.

4. *Adding or subtracting a multiple of one row from another, leaves the determine unchanged.*
   More generally, the determinant stays invariant, if one adds to a row, a linear combination of other rows (note, the remaining rows are left unchanged).

5. *The determinant of a diagonal matrix or a triangular matrix is equal to the product of the diagonal entries.*

Proof. Write

$$A = \begin{bmatrix} — & v_1 & — \\ & \vdots & \\ — & v_n & — \end{bmatrix}$$

and proceed as follows.

1. *If any two rows of A are equal, then* $\det A = 0$.
   Suppose $v_j = v_k$. Then

$$\begin{aligned} \det(A) = \quad & \det(v_1, \ldots, \underbrace{v_j}_{j\text{th row}}, \ldots, \underbrace{v_k}_{k\text{th row}}, \ldots v_n) \\ &= -\det(v_1, \ldots, \underbrace{v_k}_{j\text{th row}}, \ldots, \underbrace{v_j}_{k\text{th row}}, \ldots v_n) && \text{By Axiom 2} \\ &= -\det(A) && \because\ v_j = v_k \end{aligned}$$

   and that in turn shows $\det(A) = 0$.

2. *If A has a zero row, then* $\det A = 0$.
   Suppose $v_k = (0 \ldots 0)$ is the all-zero row. Then

$$\begin{aligned} \det(A) &= \det(v_1, \ldots, v_k, \ldots v_n) \\ &= \det(v_1, \ldots, 0 \cdot v_k, \ldots v_n) && \because\ 0 \cdot v_k = v_k \\ &= 0 \cdot \det(v_1, \ldots, v_k, \ldots v_n) && \text{By Axiom 3} \\ &= 0. \end{aligned}$$

3. *If the rows of A are linearly dependent, then* $\det A = 0$.

Without loss of generality, let $v_1 = \sum_{i=2}^{n} c_i v_i$. One can therefore write

$$\begin{aligned}
\det(A) &= \det(v_1, v_2, \dots v_n) \\
&= \det\left(\sum_{i=2}^{n} c_i v_i, v_2, \dots v_n\right) \\
&= \sum_{i=2}^{n} c_i \cdot \det(v_i, v_2, \dots v_n) && \text{By Axiom 3} \\
&= \sum_{i=2}^{n} c_i \cdot 0 && \text{From Property 1} \\
&= 0.
\end{aligned}$$

4. *Adding or subtracting a multiple of one row from another, leaves the determine unchanged.*

   The determinant of A can be written as

$$\det(A) = \det(v_1, \dots, v_j, \dots v_n).$$

   Consider the determinant of A' which is the same as A except the jth row is replaced by $v_j + cv_k$, i.e.

$$\begin{aligned}
\det(A') &= \det(v_1, \dots, \underbrace{v_j + cv_k}_{j\text{th row}}, \dots v_n) \\
&= \det(v_1, \dots, \underbrace{v_j}_{j\text{th row}}, \dots v_n) + \\
&\quad c\det(v_1, \dots, \underbrace{v_k}_{j\text{th row}}, \dots v_n) && \text{this term is zero from Property 1} \\
&= \det(A).
\end{aligned}$$

   More generally, the determinant stays invariant, if one adds to a row, a linear combination of other rows (note, the remaining rows are left unchanged).

Proceeding similarly, one has

$$\begin{aligned}
\begin{vmatrix} — & v_1 & — \\ & \vdots & \\ — & v_k + \sum_{i\neq j} c_i v_i & — \\ & \vdots & \\ — & v_n & — \end{vmatrix} &= \det\left(v_1, \ldots, \underbrace{v_j + \sum_{i\neq j} c_i v_i}_{j\text{th row}}, \ldots v_n\right) \\
&= \det\left(v_1, \ldots, \underbrace{v_j}_{j\text{th row}}, \ldots v_n\right) + \\
&\quad \sum_{i\neq j} c_i \det\left(v_1, \ldots, \underbrace{v_i}_{j\text{th row}}, \ldots v_n\right) \qquad \text{these are zero from Property 1} \\
&= \begin{vmatrix} — & v_1 & — \\ & \vdots & \\ — & v_k & — \\ & \vdots & \\ — & v_n & — \end{vmatrix} \\
&= \det(A).
\end{aligned}$$

5. *We split this into two parts*

   (a) *The determinant of a diagonal matrix is equal to the product of the diagonal entries.*

   We have

$$
\begin{aligned}
\begin{vmatrix} a_{11} & & & \\ & a_{22} & & \\ & & \ddots & \\ & & & a_{nn} \end{vmatrix} &= a_{11} \begin{vmatrix} 1 & & & \\ & a_{22} & & \\ & & \ddots & \\ & & & a_{nn} \end{vmatrix} && \text{Axiom 3} \\
&= a_{11}a_{22} \begin{vmatrix} 1 & & & & \\ & 1 & & & \\ & & a_{33} & & \\ & & & \ddots & \\ & & & & a_{nn} \end{vmatrix} \\
&= a_{11}a_{22}\dots a_{nn} \begin{vmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{vmatrix} \\
&= a_{11}a_{22}\dots a_{nn} && \text{Axiom 1.}
\end{aligned}
$$

(b) *The determinant of a triangular matrix is equal to the product of the diagonal entries.*

Suppose A is upper triangular, i.e.

$$
A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn} \end{bmatrix}
$$

and that $a_{ii} > 0$ for all $i \in \{1, \dots n\}$. One can turn A into a diagonal matrix, with diagonal entries $(a_{11}, a_{22}, \dots a_{nn})$ by taking linear combinations of its rows. For instance, to make a_{12} zero, one can subtract from the first row v_1, $a_{12} \cdot a_{22}^{-1}$ times the second row v_2. Let

$$
A' = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn} \end{bmatrix}
$$

and note that

$$
\begin{aligned}
\det A &= \det A' && \text{using Property 4} \\
&= a_{11}a_{22}\dots a_{nn} && \text{using Property 5(a) above.}
\end{aligned}
$$

Suppose A is such that $a_{ii} = 0$ for some i. It is not hard to see that one can obtain, by a linear combination of rows, a matrix that A' such that one of its rows is zero. (For instance, suppose in a 4×4

matrix, the first and the third diagonal entries are zero. Then, it is not hard to see that the third row can be made zero, e.g.

$$\begin{vmatrix} 0 & a_{12} & a_{13} & a_{14} \\ & a_{22} & a_{23} & a_{24} \\ & & 0 & a_{34} \\ & & & a_{44} \end{vmatrix} = \begin{vmatrix} 0 & 0 & a'_{13} & 0 \\ & a_{22} & a'_{23} & 0 \\ & & 0 & 0 \\ & & & a_{44} \end{vmatrix} = 0.$$

This approach readily generalises—look at the row corresponding to the last diagonal entry that is zero and one can make this row zero.) In this case,

$$\begin{aligned} \det A &= \det A' && \text{Property 4} \\ &= 0 && \text{Property 2} \\ &= a_{11}a_{22}\cdots a_{nn} && \because\ a_{ii} = 0. \end{aligned}$$

When A is lower triangular, one can proceed analogously.

□

We now turn our attention to the task of deriving an explicit formula for the determinant, starting from the three axioms in Definition 1. We look at the 2×2 and 3×3 cases first, and then derive the formula for the general case.[1]

1.3 Uniqueness from the axiomatic definition

Example 3. Show that $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$, starting from Definition 1.

[1] These examples are not strictly needed and one may choose to skip directly to Theorem 5 as well.

Solution. We have

$$
\begin{aligned}
\begin{vmatrix} a & b \\ c & d \end{vmatrix} &= \begin{vmatrix} a & 0 \\ c & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & d \end{vmatrix} && \text{Using Axiom 3 on row 1} \\
&= \underbrace{\begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix} + \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix}}_{\text{Using Axiom 3 on row 2}} + \begin{vmatrix} 0 & b \\ c & d \end{vmatrix} \\
&= \begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix} + \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \underbrace{\begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} + \begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix}}_{\text{Using Axiom 3 on row 2}} \\
&= \underbrace{\begin{vmatrix} a & 0 \\ 0 & 0 \end{vmatrix}}_{\text{subtracting } ca^{-1} \text{ row 1 from row 2}} + \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} && \text{Using Property 4} \\
&+ \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} + \underbrace{\begin{vmatrix} 0 & b \\ 0 & 0 \end{vmatrix}}_{\text{subtracting } db^{-1} \text{ row 1 from row 2}} && (1) \\
&= \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} && \text{Using Property 2} \\
&= ad - \begin{vmatrix} b & 0 \\ 0 & c \end{vmatrix} && \text{Using Property 5 and Axiom 2} \\
&= ad - bc.
\end{aligned}
$$

One can already see a pattern emerge. Every row, splits into n 'coordinate directions'. Thus, if one expands as we did above (until Equation (1)) for an $n \times n$ matrix, the expansion would contain

$$\underbrace{n \cdot n \cdot \cdots \cdot n}_{n\text{-many}} = n^n$$

terms. Most of these terms (except $n!$, as we will see), will become zero. This is because most of these terms would have 'two rows point in the same direction', i.e. have non-zero values in the same column (and zeros in the remaining columns), e.g. it will have terms like

$$\begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix}, \begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix}$$

both of which are zero (as explained above).

Let us work out the 3×3 case to build some more intuition, before we tackle the general case.

Example 4. Show that

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}\left(a_{22}a_{33} - a_{23}a_{32}\right) + a_{12}\left(a_{23}a_{31} - a_{21}a_{33}\right) + a_{13}\left(a_{21}a_{32} - a_{22}a_{31}\right),$$

starting from Definition 1.

Solution. Our strategy would again be to decompose, in terms of linear combinations of rows, one row at a time, as we did in Example 3.

- We have

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} 0 & a_{12} & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

  where each term is further decomposed into 9 more terms (3 for each of the remaining rows).

- Writing out the relevant terms in this expansion, one can continue as follows

$$= \begin{vmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & 0 & 0 \\ 0 & 0 & a_{23} \\ 0 & a_{32} & 0 \end{vmatrix} + \begin{vmatrix} 0 & a_{12} & 0 \\ a_{21} & 0 & 0 \\ 0 & 0 & a_{33} \end{vmatrix}$$
$$\begin{vmatrix} 0 & a_{12} & 0 \\ 0 & 0 & a_{23} \\ a_{31} & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_{13} \\ a_{21} & 0 & 0 \\ 0 & a_{32} & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_{13} \\ 0 & a_{22} & 0 \\ a_{31} & 0 & 0 \end{vmatrix}$$
$$\underbrace{\begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & 0 & 0 \\ a_{31} & 0 & 0 \end{vmatrix}}_{=0} + \underbrace{\begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & 0 & 0 \\ 0 & a_{32} & 0 \end{vmatrix}}_{=0} + \underbrace{\cdots}_{\text{(all zeros)}}.$$

- All but 6 determinants are zero because in all other terms, at least one 'row direction' is repeated.
  - These 6 non-zero determinants, as we now justify, actually correspond to 3! choices.
    - There are $n = 3$ choices for the first row.
    - Once that is fixed, one cannot have a term in that column, so there are $n - 1 = 2$ choices for the next row.
    - And finally, there is only 1 choice for the last row.

- Focussing on the non-zero terms, we have

$$
\begin{aligned}
\det A = & \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \\
= & a_{11}a_{22}a_{33} \begin{vmatrix} 1 & & \\ & 1 & \\ & & 1 \end{vmatrix} + a_{11}a_{23}a_{32} \begin{vmatrix} 1 & & \\ & & 1 \\ & 1 & \end{vmatrix} \\
& a_{12}a_{21}a_{33} \begin{vmatrix} & 1 & \\ 1 & & \\ & & 1 \end{vmatrix} + a_{12}a_{23}a_{31} \begin{vmatrix} & 1 & \\ & & 1 \\ 1 & & \end{vmatrix} + \\
& a_{13}a_{21}a_{32} \begin{vmatrix} & & 1 \\ 1 & & \\ & 1 & \end{vmatrix} + a_{13}a_{22}a_{31} \begin{vmatrix} & & 1 \\ & 1 & \\ 1 & & \end{vmatrix} \\
= & a_{11}\left(a_{22}a_{33} - a_{23}a_{32}\right) + a_{12}\left(a_{23}a_{31} - a_{21}a_{33}\right) + \\
& a_{13}\left(a_{21}a_{32} - a_{22}a_{31}\right) \qquad (2)
\end{aligned}
$$

  as asserted.

- We now write this expression more systematically, to see if a more generalisable formula can be obtained.
  - First, let us recall some basics of permutation.
    - Notation.
      - An example of a permutation σ on $\{1,2,3\}$ is $\sigma = (1,3,2)$ which means $\sigma(1) = 1, \sigma(2) = 3, \sigma(3) = 2$.
      - We define the sign of a permutation as

$$\mathsf{sign}(\sigma) = \begin{cases} 1 & \text{if } N_\sigma \text{ is even} \\ 0 & \text{if } N_\sigma \text{ is odd} \end{cases}$$

        with N_σ being the number of pairwise exchanges required to transform σ to $(1,2,3)$.
    - Fact.
      - Every permutation of a finite set can be expressed as the product of transpositions.
      - For a given permutation, although many such expressions can exist
        they all contain either
        an even number of transpositions or
        an odd number of transpositions.
  - Returning to our discussion, note that one can continue with Equation (2) above and write

$$\det(A) = \sum_{\sigma \in S_3} a_{1\sigma(1)} a_{2\sigma(2)} a_{3\sigma(3)} \mathsf{sign}(\sigma) \qquad (3)$$

where

$$\begin{aligned} S_3 :=& \text{Set of all permutations on } \{1,2,3\} \\ =& \{(1,2,3),(1,3,2),(2,1,3),(2,3,1),(3,1,2),(3,2,1)\}\,. \end{aligned}$$

Our expression for the determinant for the 3×3 case Equation (3) is quite suggestive.

We now use our intuition to work out the general expression for the determinant of an $n \times n$ matrix.

Theorem 5. *Let* det *be any function satisfying Definition 1. Let* $A \in \mathbb{F}^{n\times n}$*. Then* $\det(A)$ *is uniquely specified by*

$$\det(A) = \sum_{\sigma \in S_n} a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)} \mathsf{sign}(\sigma)$$

where S_n *is the set of all permutations on* n *elements.*

Proof. The strategy again is to

- decompose each row such that it has only one non-zero column vector,
  - by invoking Axiom 3 repeatedly, and
- argue that all but $n!$ terms yield 0
  - by using the various properties of the determinant we proved (primarily Property 1).

Denote by a_{ij} the ith row and jth column of

$$A = \begin{bmatrix} — & v_1 & — \\ & \vdots & \\ — & v_n & — \end{bmatrix}.$$

One can write

$$\begin{aligned} v_i &= (a_{i1}, \ldots a_{in}) \\ &= \sum_{j=1}^{n} a_{ij} e_j \end{aligned}$$

where $e_1, \dots e_n$ is the standard basis. Now, one can further write

$$\begin{aligned}
\det A &= \det(v_1, v_2 \dots v_n) \\
&= \det\left(\sum_{j=1}^{n} a_{1j} e_j, v_2, \dots v_n\right) \\
&= \sum_{j=1}^{n} a_{1j} \det\left(e_j, v_2, \dots v_n\right) && \text{Axiom 3} \\
&= \sum_{j_1=1}^{n} \sum_{j_2=1}^{n} a_{1j_1} a_{2j_2} \det(e_{j_1}, e_{j_2}, v_3, \dots v_n) && \text{Axiom 3} \\
&= \sum_{j_1=1}^{n} \sum_{j_2=1}^{n} \cdots \sum_{j_n=1}^{n} a_{1j_1} a_{2j_2} \dots j_{nj_n} \det(e_{j_1}, \dots e_{j_n}) && \text{Axiom 3.}
\end{aligned}$$

- Note that this sum contains

$$\underbrace{n \cdot n \cdot \dots \cdot n}_{n\text{- many}} = n^n$$

  terms but most of these are zero.

  - Concretely, if any two of the indices $j_1, \dots j_n$ coincide, then $\det(e_{j_1}, \dots e_{j_n}) = 0$ (by Property 1).
  - Consequently, the sum can be simplified as follows

$$\det A = \sum_{\sigma \in S_n} a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)} \det(e_{1\sigma(1)}, \cdots, e_{n\sigma(n)}) \tag{4}$$

    where S_n is the set of all permutations on $\{1, 2 \dots, n\}$ and $|S_n| = n!$.

- Note also that

$$\det(e_{1\sigma(1)}, \dots e_{n\sigma(n)}) = \mathsf{sign}(\sigma) \tag{5}$$

  using Axiom 2 and Axiom 1.

Combining Equations (4) and (5), we get the desired expression. □

This completes the main result we wanted to discuss. We now look at some other properties of the determinant.

Example 6. Show that $\det(A^T) = \det(A)$ where A^T is the transpose of A, i.e. $(A^T)_{ij} = (A)_{ji}$ for all i, j.

Solution. Let $B = A^T$. This means that $b_{ij} = a_{ji}$ for all i (we are using b_{ij} and a_{ij} to denote the ith row and jth column of B and A respectively). Note

$$\begin{aligned}
\det A^T = \det B &= \sum_{\sigma \in S_n} b_{1\sigma(1)} b_{2\sigma(2)} \cdots b_{n\sigma(n)} \mathsf{sign}(\sigma) \\
&= \sum_{\sigma \in S_n} a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n} \mathsf{sign}(\sigma) \\
&= \sum_{\sigma \in S_n} a_{1\sigma^{-1}(1)} a_{2\sigma^{-1}(2)} \cdots a_{n\sigma^{-1}(n)} \mathsf{sign}(\sigma) \\
&= \sum_{\pi \in S_n} a_{1\pi(1)} a_{2\pi(2)} \cdots a_{n\pi(n)} \mathsf{sign}(\pi^{-1}) && \text{take } \pi = \sigma^{-1} \\
&= \sum_{\pi \in S_n} a_{1\pi(1)} a_{2\pi(2)} \cdots a_{n\pi(n)} \mathsf{sign}(\pi) && \because \mathsf{sign}(\pi) = \mathsf{sign}(\pi^{-1}) \\
&= \det A.
\end{aligned}$$

Proposition 7. *It holds that* $\det(AB) = \det A \cdot \det B$ *if* $\det A \neq 0$ *or* $\det B \neq 0$.

Proof. Proceed as follows.

- Define $d(A, B) := \det(AB)/\det(B)$. Treat d as a function of A (keep B fixed).
- Claim (left as an exercise). $d(A)$ satisfies the axioms in Definition 1 (i.e. $d(I) = 1$, anti-symmetry and linearity in any row).
- We already showed (in Theorem 5) that any function that satisfies the axioms in Definition 1 is uniquely specified, we conclude that

$$d(A) = \det(AB)/\det(B) = \det(A).$$

□

2 Recommended reading (ungraded)

In this part, we briefly look at the trace and a few applications of the determinant.

2.1 Trace

Definition 8 (Trace)**.** For an $n \times n$ matrix A, the trace of A is defined as $\mathsf{Tr}(A) = \sum_{i=1}^{n} a_{ii}$.

- The trace is very useful in a variety of situations. Here's a nice relation between determinants, traces and the exponentiation function, that is extended to matrices by its power-series[2]

$$\det(\exp(A)) = \exp(\mathsf{Tr}(A)).$$

[2] $\exp(A) := 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \cdots$

This also shows up in physics.

2.2 Inverse

We outline the idea behind the proof of this statement.

- $\det A$ turns out to be useful for computation of the inverse of a matrix, if it exists.
- For the case of 2×2 matrices, since we already have the formula for inverse, we can try to see how it relates to the determinant, i.e.

$$\begin{aligned}\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} &= \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \\ &= \frac{1}{\det A}\begin{bmatrix} c_{11} & c_{21} \\ c_{12} & c_{22} \end{bmatrix}\end{aligned}$$

  where c_{ij} is the co-factor of (i,j)th entry defined as $c_{ij} = (-1)^{i+j}\det M_{ij}$,

  - where M_{ij} is the sub-matrix of A obtained by removing the ith row and jth column.

- In general, it turns out that one can compute inverse of A using

$$\boxed{A^{-1} = \frac{C^T}{\det A}}$$

  where C is the matrix with c_{ij} as its entry in the ith row and jth column.

  - Note that the equation above says $(A^{-1})_{ij} = c_{ji}/\det A$.

Fact 9. $AA^{-1} = A^{-1}A = I$.

Proof. Suppose the left and right inverses were different, call them L and R. Then

$$L = LI = L(AR) = (LA)R = IR = R,$$

i.e. the left and right inverses must be the same. □

Lemma 10. *Let A be an $n \times n$ matrix. Then, A is invertible if and only if* $\det A \neq 0$.

Proof sketch. To see that $\det A \neq 0$ is necessary for the inverse to exist, suppose A^{-1} exists. Then

$$\begin{aligned}\det(AA^{-1}) &= \det(A)\det(A^{-1}) \\ &= \det(I) = 1\end{aligned}$$

since[3] $\det(AA^{-1}) = \det(I)$. This equation can only hold if both $\det A$ and $\det A^{-1}$ are non-zero.

To see why this is sufficient, we invoke the formula for A^{-1} above. □

[3] Not very happy with this because we proved $\det(AB) = \det A \cdot \det B$ only holds when at least one of $\det A, \det B$ is non-zero. But the formula, it appears, hold in general.

2.3 Relation with eigenvalues

To state the next lemma, we recall that

- λ is an eigenvalue of an $n \times n$ matrix A if

$$Ax = \lambda x, \quad \forall x \in \mathbb{F}^n, x \neq 0$$

  and $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

- λ is an eigenvalue of an operator T if there is a non-zero x such that $Tx = \lambda x$.

Lemma 11. *λ is an eigenvalue of a matrix A if and only if* $\det(A - \lambda I) = 0$.

Proof. Note that for an $x \neq 0$,

$$\begin{aligned} & Ax = \lambda x \\ \iff & (A - \lambda I)x = 0, \\ \iff & (A - \lambda I) \qquad\qquad \text{is not invertible.} \end{aligned}$$

Using Lemma 10 we have that A' is invertible if and only if $\det A' \neq 0$. Setting $A' = (A - \lambda I)$, we obtain the result. □

Remark 12. For an $n \times n$ matrix A, the roots of $\det(A - \lambda I) = 0$ gives the eigenvalues of A. These are the n eigenvalues counting multiplicities of the roots.

We are almost done. Let us look at how the trace and determinant relate to eigenvalues of a matrix.

Theorem 13. *Let A be an $n \times n$ matrix over complex numbers $\mathbb{C}$ and let $\lambda_1, \ldots \lambda_n$ be its eigenvalues (counting multiplicities). Then, $\mathsf{Tr}(A) = \sum_{i=1}^n \lambda_i$ and $\det A = \prod_{i=1}^n \lambda_i$.*

Proof. We start with $\det A$. We have

$$\det(A - \lambda I) = \prod_{i=1}^{n} (\lambda_i - \lambda)$$

from Remark 12 for any $\lambda \in \mathbb{C}$. Setting $\lambda = 0$, we obtain $\det A = \prod_{i=1}^n \lambda_i$.

Let us now look at $\mathsf{Tr}(A)$. The idea here is to look at $\det(A - \lambda I)$ as a polynomial in λ but in two different ways and then equating the appropriate coefficient of λ^{n-1}.

- Observe that the coefficient of λ^{n-1} in $\det(A - \lambda I) = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda)$ is

$$\mathsf{coeff} := (-1)^{n-1}(\lambda_1 + \cdots + \lambda_n). \tag{6}$$

- Also, observe that

$$\det(A - \lambda I) = \sum_{i=1}^{n}(a_{ii} - \lambda) + q(\lambda)$$

  where $q(\lambda)$ is a polynomial of degree at most $n-2$ (in λ).

  - This is easily seen from the formula of det in Theorem 5 and
  - noting that the remaining terms will have at least two terms that are not along the diagonal.
  - Therefore, the coefficient of λ^{n-1} in $\det(A - \lambda I)$ must come from
    - the coefficient of λ^{n-1} $\sum_{i=1}^{n}(a_{ii} - \lambda)$,
    - which in turn is

$$\mathsf{coeff}' := (-1)^{n-1}(a_{11} + \cdots + a_{nn}) = (-1)^{n-1}\mathsf{Tr}(A).$$

- Clearly, coeff and coeff' are the same quantity, evaluated differently. Equating them, we get

$$\mathsf{Tr}(A) = (\lambda_1 + \cdots + \lambda_n)$$

  as asserted.

□

2.4 Determinants and Traces of operators

We end this discussion by extending the notion of trace and determinants to operators. We start with trace.

Remark 14. In Problem 1 in Problem Set 9, we had stated the following:

> λ is an eigenvalue of an operator $T \iff$ the matrix $[T - \lambda I]_B$ is not invertible for any basis B.

Thus, the eigenvalues of an arbitrary operator $T \in L(V)$ can be computed by fixing any basis B and solving $\det([T - \lambda I]_B) = 0$.

One might wonder whether the result could on the choice of the basis. It cannot.

Claim 15. Prove that $\det([T - \lambda I]_B) = \det([T - \lambda I]_{B'})$ where B, B' are distinct bases.

Proof sketch.

- Show that $[T - \lambda I]_B$ and $[T - \lambda I]_{B'}$ are *similar* matrices.
- Use that for any matrix T', $[T']_B$ is similar to $[T']_{B'}$.
- Then argue that similar matrices have the same determinant.

□

We are now ready to define trace and determinant for operators.

- Based on this discussion, and the fact that $\mathsf{Tr}(A)$ for a matrix is simply the sum of its eigenvalues (see Theorem 13), one can define trace for an operator T as follows:
  - For $T \in \mathcal{L}(V)$,

$$\begin{aligned}\mathsf{Tr}(T) :=& \text{sum of eigenvalues of } T\\ =& \text{sum of eigenvalues of } [T]_B\\ =& \lambda_1 + \lambda_2 + \cdots + \lambda_n.\end{aligned}$$

- One can proceed similarly for the determinant, i.e.
  - note that it stays invariant under similarity transformations and
  - use the characterisation as product of eigenvalues from Theorem 13, to write
  - For $T \in \mathcal{L}(V)$,

$$\begin{aligned}\det(A) :=& \text{product of eigenvalues of } T\\ =& \text{product of eigenvalues of } [T]_B\\ =& \lambda_1 \cdots \lambda_n.\end{aligned}$$